

|                                                    |                                          |
|----------------------------------------------------|------------------------------------------|
| Program: <b>Diploma in Automation and Robotics</b> |                                          |
| Course Code: <b>5338</b>                           | Course Title: <b>AI for Robotics Lab</b> |
| Semester: <b>5</b>                                 | Credits: <b>3</b>                        |
| Course Category: <b>Program core</b>               |                                          |
| Periods per week: <b>3 (L:0 T:0 P:3)</b>           | Periods per semester: <b>45</b>          |

### Course Objectives:

- To introduce hand on experience on Artificial Intelligence problems.
- To get practical knowledge on implementing machine learning algorithms in real time problem for getting solutions

### General Instructions:

- Students have to write the date, aim, Software & Hardware requirements for that experiment in the observation book.
- Students have to listen and understand the experiment explained by the faculty and note down the important points in the observation book.
- Students need to write procedure/algorithm in the observation book.
- Analyze and Develop/implement the logic of the program by the student in respective platform
- After approval of logic of the experiment by the faculty then the experiment has to be executed on the system.
- After successful execution the results are to be shown to the faculty and noted the same in the observation book.
- Students need to attend the Viva-Voce on that experiment and write the same in the observation book.
- Update the completed experiment in the record and submit to the concerned faculty in-charge.

### Course Prerequisites:

| Topic                              | Course code | Coursename              | Semester |
|------------------------------------|-------------|-------------------------|----------|
| Basics of Arduino, Microcontroller | 2371        | Foundations of Robotics | 2        |

**Tools and Equipment:**

Processors: Intel Atom® processor or Intel® Core™ i3 processor.

Disk space: 1 GB.

Operating systems: Windows 7 or later, macOS, and Linux.

Software : PyCharm, Arduino IDE

**Course Outcomes:**

On completion of the course, the student will be able to:

| Con | Description                                                           | Duration (Hours) | Cognitivelevel |
|-----|-----------------------------------------------------------------------|------------------|----------------|
| CO1 | To Understand the concept of Artificial intelligence                  | 6                | Applying       |
| CO2 | To apply various search algorithms of artificial intelligence.        | 12               | Applying       |
| CO3 | To apply knowledge representation and reasoning techniques.           | 6                | Applying       |
| CO4 | To understand & apply different types of machine learning and models. | 18               | Applying       |
|     | Lab Exam                                                              | 3                |                |

**CO–PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     |     | 3   |     |     |
| CO2             | 3   |     |     |     |     |     |     |
| CO3             | 3   |     |     | 3   |     |     |     |
| CO4             | 3   |     |     | 3   |     |     |     |
| CO5             | 3   |     |     | 3   |     |     |     |

3-Stronglymapped,2-Moderatelymapped,1-Weaklymapped

**Course Outline:**

| Module Outcomes | Description                                                     | Duration (Hours) | Cognitive Level |
|-----------------|-----------------------------------------------------------------|------------------|-----------------|
| CO1             | <b>To Understand the concept of Artificial intelligence.</b>    |                  |                 |
| M1.01           | Write a program in prolog to implement simple facts and Queries | 3                | Applying        |

|            |                                                                                  |     |          |
|------------|----------------------------------------------------------------------------------|-----|----------|
| M1.02      | Write a program in prolog to implement simple arithmetic                         | 3   | Applying |
| <b>CO2</b> | <b>To apply various search algorithms of artificial intelligence</b>             |     |          |
| M2.01      | Write a program in prolog to solve Tower of Hanoi                                | 3   | Applying |
| M2.02      | Write a program in prolog to solve 8 Puzzle problems                             | 3   | Applying |
| M2.03      | Write a program in prolog to implement Breadth First Search Traversal            | 3   | Applying |
| M2.04      | Write a program in prolog to implement Water Jug Problem                         | 3   | Applying |
|            | Lab Exam I                                                                       | 1.5 |          |
| <b>CO3</b> | <b>To apply knowledge representation and reasoning techniques.</b>               |     |          |
| M3.01      | Write a program in prolog to solve 4-Queens problem                              | 3   | Applying |
| M3.02      | Implementation of A* Algorithm using PROLOG                                      | 3   | Applying |
| <b>CO4</b> | <b>To understand &amp; apply different types of machine learning and models.</b> |     |          |
| M4.01      | Implementation of Python libraries such as numpy, matplotlib, Panda              | 3   | Applying |
| M4.02      | Extract the data from database using python                                      | 3   | Applying |
| M4.03      | Implement k-nearest neighbours classification using python                       | 4   |          |
| M4.04      | Implement linear regression using python.                                        | 4   |          |
| M4.05      | Detecting Spam mails using Support vector machine                                | 4   |          |
|            | Lab Exam II                                                                      | 1.5 |          |

#### Online Resources:

| Sl No | Website Link/Text books                               |
|-------|-------------------------------------------------------|
| 1     | <a href="https://github.com/">https://github.com/</a> |